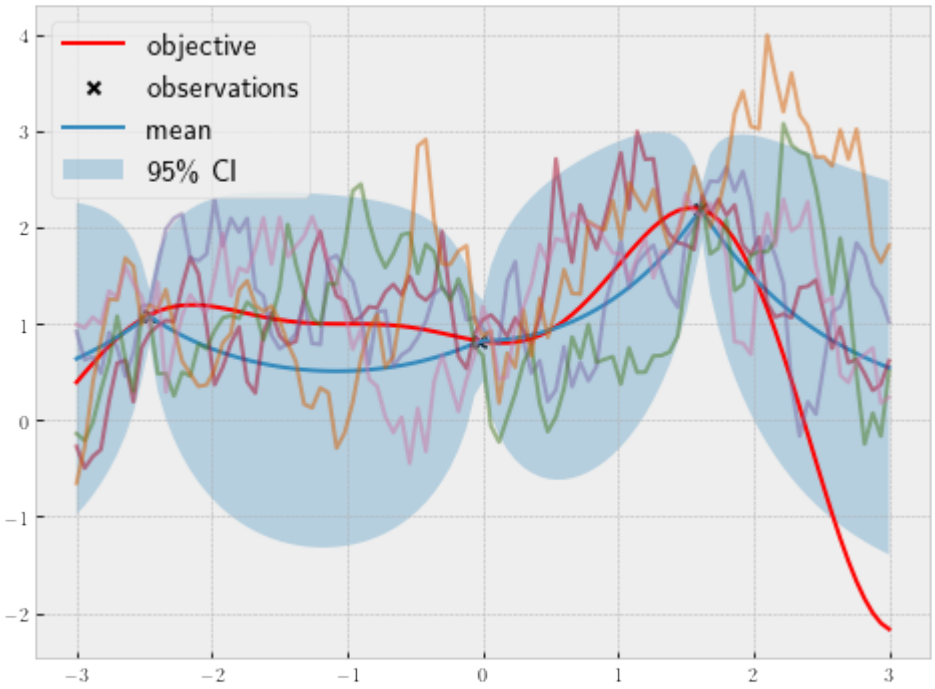


Figure 3.13 GP's predictions with the Matérn 0.5 kernel, which induces the belief that the objective function is not differentiable, corresponding to very rough samples.

Unlike what we have seen before with RBF, the samples from this Matérn kernel are all very jagged. In fact, none of them are differentiable. The value `nu = 0.5` is a good value for the Matérn kernel when modeling time-series data such as stock prices.

However, this value is typically not used in Bayesian optimization, as jagged functions like those in Figure 3.13 are highly volatile (they can jump up and down in an unpredictable way) and are usually not the target for automated optimization techniques. We need a certain level of smoothness from our objective function that is to be optimized; otherwise, effective optimization is an unrealistic goal.

The Matérn 2.5 kernel is commonly preferred. Its predictions, along with those generated by Matérn 1.5, are visualized in Figure 3.14.

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